Enterprise AI & Microsoft Technology Training

Microsoft Copilot Studio

Day 1 — Foundations and Getting Started

Participant Handbook

Handbook detail	Value
Course	Microsoft Copilot Studio — Instructor-Led Training (ILT)
Day / Duration	Day 1 of the series — 1 full day (approx. 7 hours + breaks)
Delivery mode	Instructor-led, hands-on (classroom or virtual)
Audience	Business analysts, functional consultants, IT pros, citizen developers, new makers
Modules covered	Module 1, Module 2, Module 3 + Labs 1 to 6
Product version	Microsoft Copilot Studio — classic experience and the new agent experience (production-ready preview)
Document version	v1.0 — July 2026
Prepared by	Learning Team

How to Use This Handbook

This handbook is your companion for the whole day. You do not need to write everything down — the explanations, the steps and the links are already here. Use the white space in the margins for your own notes.

Everything is written in plain language. If you have never used Power Platform before, you will still be able to follow along. Technical terms are explained the first time they appear and again in the Glossary (Appendix A).

Icons and boxes you will see

Box	What it means
NOTE	Important background information. Read it, but you do not have to act on it.
TIP	A shortcut or a trick from the field that saves you time.
WATCH OUT	A common mistake. Read this before you click.
BEST PRACTICE	How experienced makers do it in real projects.
[SCREENSHOT x.x]	A placeholder where your trainer will show, or your printed copy will carry, the matching product screen.

> **NOTE**
> Copilot Studio ships updates every month. Screens in your environment may differ slightly from what your trainer shows. The concepts stay the same. When in doubt, the Microsoft Learn links at the end of each module are the source of truth.

Structure of every module

- **Learning Objectives** — what you will be able to do at the end.
- **Detailed Concepts** — the explanation, in simple language, with diagrams and screenshots.
- **Step-by-step Activity** — a short guided walkthrough you do with the trainer.
- **Best Practices and Common Mistakes** — field experience distilled into a checklist.
- **Knowledge Check** — a few questions to test yourself. Answers are in Appendix C.
- **Key Takeaways** — the three or four things you must remember.
- **Learn More** — official Microsoft Learn modules and documentation links.

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NOTE
The page numbers in this table update automatically. In Microsoft Word, press Ctrl+A then F9, and choose 'Update entire table'.

Day 1 at a Glance

Day 1 takes you from 'I have heard of Copilot' to 'I have built, tested and published my own working agent'. You will spend roughly half the day on concepts and half the day in the product.

Agenda

Time	Session	Format
09:00 – 09:30	Welcome, introductions, course objectives	Discussion
09:30 – 11:00	Module 1 — Introduction to Copilot Studio	Concept
11:00 – 11:15	Break	—
11:15 – 12:15	Lab 1 — Create Your First Agent · Lab 2 — Explore the Interface	Hands-on
12:15 – 13:00	Module 2 — Environment Setup	Concept
13:00 – 14:00	Lunch	—
14:00 – 15:00	Lab 3 — Configure Environment and Security Roles	Hands-on
15:00 – 15:45	Module 3 — Build Your First Copilot	Concept
15:45 – 16:00	Break	—
16:00 – 17:00	Lab 4 — Instructions · Lab 5 — Test Canvas · Lab 6 — Publish	Hands-on
17:00 – 17:30	Recap, knowledge check review, Day 2 preview	Discussion

What you need before you start

Requirement	Details
Browser	Microsoft Edge or Google Chrome, current version. Copilot Studio is a web application at copilotstudio.microsoft.com.
Work or school account	A Microsoft Entra ID (work) account. Personal Microsoft accounts are not supported.
Licence or trial	A Copilot Studio user licence, a Microsoft 365 Copilot licence, a Copilot Studio trial, or membership of the Copilot Studio authors security group. See Module 1, section 1.3.
Environment access	Access to a Power Platform environment that has a Dataverse database, with the Environment Maker security role.
Optional	A sample PDF or a public website URL you can use as a knowledge source in Lab 4.

WATCH OUT

The free Copilot Studio trial licence lets you create and test agents, but it does not let you publish them. If you want to complete Lab 6 end to end, ask your administrator for a full licence or use the classroom tenant provided by your trainer.

Module 1 — Introduction to Copilot Studio

Duration: approximately 90 minutes. Related labs: Lab 1, Lab 2.

Learning Objectives

By the end of this module you will be able to:

- Describe how Microsoft's AI assistants evolved from chatbots into agents, and explain why the word 'agent' matters.
- Explain in one sentence what Copilot Studio is and who it is for.
- List the ways an organisation can license Copilot Studio and check what you personally need to get started.
- Draw the high-level architecture of Copilot Studio and name each layer.
- Name the main components of an agent — instructions, knowledge, tools, and channels — and say what each one does.
- Walk through the five stages of the AI agent lifecycle and say what 'done' looks like at each stage.

1.1 Evolution of Microsoft AI Agents

It helps to know where this product came from, because you will still meet the older names in menus, documentation and in your organisation's existing projects.

From scripted bots to reasoning agents

Era	What it was called	How it worked	What it could not do
Around 2019	Power Virtual Agents	You drew conversation trees by hand. Every question, every answer and every branch was designed by a person.	Anything the designer had not thought of. Unexpected wording broke the conversation.
2023	Copilot	Large language models were added to Microsoft 365 apps. The assistant could summarise, draft and answer in natural language.	Act on your systems. It could talk, but not do.
2023 – 2024	Copilot Studio (renamed from Power Virtual Agents)	Makers could combine designed conversation topics with generative answers over their own documents.	Plan a multi-step task on its own.
2024 – 2025	Agents	Generative orchestration arrived. The agent chooses which topic, knowledge source or tool to use, and can run without a human starting the conversation (autonomous triggers).	Reliably chain many steps across systems.
2025 – 2026	Agent systems and the new agent experience	You describe the agent in plain English. An enhanced orchestration runtime reasons over instructions, knowledge, tools and skills. Agents can call other agents, run workflows, and even drive application user interfaces.	—

> **NOTE**
> Power Virtual Agents was renamed to Microsoft Copilot Studio. If you find an old tutorial that says 'Power Virtual Agents', it is describing the same product family — but the interface has changed a lot, so treat the steps with caution.

[SCREENSHOT 1.1]

The Copilot Studio home page, showing agent templates and the option to describe an agent in natural language.

Chatbot, copilot, agent — what is the difference?

- **Chatbot:** follows a script you wrote. Predictable, but brittle. If the user types something unexpected, it fails.

- **Copilot:** sits beside a person and helps them work — drafting, summarising, answering. The person is still driving.
- **Agent:** is given a goal, a set of instructions, some knowledge and some tools, and then decides for itself which steps to take. It can be started by a person or by an event, such as an email arriving.

> **TIP**
> A simple test in the classroom: if you can write down every possible path in advance, you want a chatbot-style topic. If you cannot, you want an agent with good instructions.

1.2 What is Copilot Studio?

Microsoft Copilot Studio is a low-code, web-based tool for building, testing, publishing and managing AI agents. You build the agent in a browser. You describe what it should do in ordinary language, point it at the information it should use, connect it to the systems it should act on, and then publish it to the places where your users already work — Microsoft Teams, the Microsoft 365 Copilot app, your website, a phone line, and more.

Who uses it

Role	What they do in Copilot Studio
Business user / subject-matter expert	Describes the agent's purpose and writes its instructions. Adds documents as knowledge. Tests the answers for accuracy.
Maker / functional consultant	Configures knowledge sources, adds tools, designs conversation topics where precise control is needed, and publishes to channels.
Pro developer	Builds custom connectors, calls REST APIs, extends agents with code, and integrates through the Direct Line API or the Microsoft 365 Agents SDK.
Administrator	Creates and governs environments, assigns licences and security roles, applies data policies, and monitors cost and usage.

Two authoring experiences you will meet

As of 2026, Copilot Studio offers two ways to build an agent. Your trainer will tell you which one your classroom tenant is using. You choose the experience when you create the agent, and an agent cannot be moved from one experience to the other.

	Classic experience	New agent experience
How you build	Topics, trigger phrases, nodes and branching logic on a canvas.	You describe the agent in natural language; instructions and reasoning drive behaviour.
Navigation	Separate left-hand items: Topics, Knowledge, Actions, Settings.	One surface with tabs across the top: Build, Preview, Evaluate, Monitor.
Orchestration	You choose classic or generative orchestration.	One enhanced orchestration runtime for every agent; not configurable.
Status	Fully supported, broadest feature set.	Production-ready preview; some classic features are not there yet.
Choose it when	You need step-by-step, deterministic control, or you are maintaining an existing agent.	You are starting fresh and want better reasoning, especially over Microsoft 365 data.

> **TIP**
> Not sure which experience you are in? Look at the left-hand navigation. If you see Topics, Knowledge, Actions and Settings as separate items, you are in the classic experience. If you see Build, Preview, Evaluate and Monitor as tabs, you are in the new experience.

[SCREENSHOT 1.2]

Side-by-side comparison: the classic agent overview page and the new experience Build page.

1.3 Licensing and Prerequisites

Licensing is the question that stops most first projects. Keep two ideas separate in your head:

- **Access** — am I allowed to open Copilot Studio and build an agent?
- **Consumption** — who pays when people actually talk to my agent?

Access — what you need to start building

You need any one of the following:

Option	What it is	Good to know
Copilot Studio user licence	Assigned to you by an admin in the Microsoft 365 admin centre. There is no per-user charge for the licence itself.	Your tenant must already have a prepaid Copilot Credit pack subscription before this user licence can be granted.
Copilot Studio authors role	Your admin creates a Microsoft Entra security group, adds you to it, and assigns that group to the Copilot Studio author setting in the Power Platform admin centre.	The usual approach in larger organisations — access is managed by group membership.
Microsoft 365 Copilot licence	Lets you use Copilot Studio to build and extend agents for Microsoft 365 Copilot.	Internal usage inside Microsoft 365 Copilot, Teams and SharePoint is zero-rated for certain answer types.
Copilot Studio trial licence	Individual sign-up, time limited.	You can create and test agents, but you cannot publish them.

Consumption — Copilot Credits

Since 1 September 2025 the common currency across Copilot Studio is the **Copilot Credit**. A credit is a measure of the time and effort an agent needs to retrieve information, respond to a prompt, and carry out actions such as running a workflow or calling a tool. A complex, multi-step answer costs more credits than a simple one. Voice agents are charged on call length, to the nearest second.

Purchasing option	How it works
Copilot Studio prepaid capacity pack (subscription)	A monthly subscription. One credit pack equals 25,000 Copilot Credits. Unused credits do not roll over to the next month.
Pay-as-you-go meter	Link a Power Platform environment to an Azure subscription with a billing policy. You pay at the end of the month for what was actually consumed. No upfront commitment.
Copilot Credit Pre-Purchase Plan (P3)	A one-year, pay-upfront pool of Copilot Credit Commit Units (CCCUs), usable across eligible Microsoft products. Unused units expire at the end of the term.
Microsoft Agent Pre-Purchase Plan (P3)	A one-year, pay-upfront pool of Agent Commit Units (ACUs). Unused units expire at the end of the term.

> **WATCH OUT**
>
> Purchased capacity is enforced monthly. If your usage goes past what you bought, technical enforcement can apply and service can be denied. Before you roll an agent out to thousands of users, estimate the traffic — Microsoft publishes an agent usage estimator for exactly this purpose.

> **BEST PRACTICE**
>
> Even if you buy a subscription capacity pack, also set up the pay-as-you-go meter as a safety net for overage months.
>
> Run the Copilot Studio agent usage estimator during design, not after go-live. Traffic volume, orchestration type, knowledge sources and tools all change the answer.
>
> Licensing terms change often. Treat the Microsoft Copilot Studio Licensing Guide PDF as the authority, and re-check it before every commercial commitment.

Technical prerequisites checklist

- A work or school account in Microsoft Entra ID. Personal accounts do not work.

- A Power Platform environment with a Dataverse database provisioned.
- The Environment Maker security role in that environment (Module 2 covers this).
- A supported modern browser. No installation is required.
- For publishing to Teams or Microsoft 365 Copilot: administrator approval in the Teams admin centre or Microsoft 365 admin centre.

1.4 Copilot Studio Architecture

You do not need to be an architect to build an agent, but knowing the layers helps enormously when something does not work — because you can tell where to look.

[SCREENSHOT 1.3]

Layered architecture diagram: channels at the top, agent runtime and orchestrator in the middle, knowledge, tools and Dataverse below, governance wrapping the whole.

Layer	What lives here	Why you care
1. Channel layer	Microsoft Teams, the Microsoft 365 Copilot app, your website, SharePoint, Power Pages, voice and telephony, messaging platforms, custom apps via Direct Line.	This is where your users meet the agent. One agent, many channels.
2. Authoring layer	The Copilot Studio maker portal in your browser: instructions, knowledge, tools, topics, settings, test canvas, analytics.	This is where you spend your day.
3. Orchestration and reasoning	The runtime that reads the user's message, decides what the user wants, and picks the right knowledge source, topic or tool to satisfy it. In the new experience this is the enhanced orchestration runtime.	When an agent gives a strange answer, this layer decided it. The fix is usually clearer instructions or better-scoped knowledge.
4. Model layer	The underlying large language models, plus model choice where available, content moderation and responsible AI controls.	Determines answer quality, cost and, in some cases, data region.
5. Knowledge layer	Uploaded files, public websites, SharePoint, Dataverse, Azure AI Search, Dynamics 365, Salesforce, ServiceNow, Azure SQL, and Microsoft IQ for Microsoft 365 organisational data.	Grounding. Without it, the agent falls back on general reasoning and is far more likely to be wrong.
6. Action and tool layer	Connectors, Power Automate flows and agent flows, custom prompts, REST APIs, MCP servers, connected agents.	This is how an agent stops talking and starts doing.
7. Data and platform layer	Microsoft Dataverse stores the agent definition, its components and its conversation transcripts, inside a Power Platform environment.	Explains why environments and security roles matter so much — see Module 2.
8. Governance layer	Power Platform admin centre, data loss prevention policies, managed environments, analytics, audit, and Microsoft Agent 365 as a central control plane for agents.	The layer that decides whether your pilot is allowed to become production.

> **NOTE**
> Every agent you create lives inside a Power Platform environment and is stored in that environment's Dataverse database. Delete the environment and you delete the agent. This is the single most important architectural fact for a beginner.

1.5 Components and Capabilities

In the new agent experience, an agent is built from five components, all configured from the Build tab. In the classic experience the same ideas exist, but they sit in separate left-hand navigation items.

Component	In plain language	Example
Instructions	Who the agent is, how it should speak, what it is allowed to do, and where it must stop.	"You are the HR assistant for Contoso India. Answer only from the HR policy documents. If you do not know, say so and share the HR helpdesk email."
Knowledge	The information the agent is allowed to read when answering. Files, websites, SharePoint, business systems, and Microsoft IQ for Microsoft 365 data such as email, calendar, files and Teams messages.	The leave policy PDF, the intranet FAQ page, a SharePoint site.
Tools and skills	What the agent can do: call an API, run a Power Automate flow, use a custom prompt. Skills are reusable, self-contained sets of instructions.	Create a ticket in ServiceNow. Look up an order in Dynamics 365.
Model	The AI model that powers the agent's reasoning, where model choice is offered.	Choosing a model for cost, region or multilingual strength.
Connected agents	Other agents that yours can hand work to, so each stays focused on what it does best.	An IT agent that delegates password resets to a dedicated identity agent.

The four tabs of the new agent experience

Tab	Purpose
Build	Configure identity, instructions, knowledge, tools, skills and model — everything in one place.
Preview	Chat with the agent while you build. This is the test canvas.
Evaluate	Create and run test sets to measure the agent's quality, not just eyeball a few answers.
Monitor	Review recent tasks, what the agent accessed, and its activity after it goes live.

[SCREENSHOT 1.4]

The Build tab of the new agent experience, showing instructions on the left and the components panel — Knowledge, Tools, Skills, Microsoft IQ — on the right.

Capabilities at a glance

- **Generative answers** grounded in your documents and systems, with citations back to the source.
- **Topics** (classic experience) for conversations that must follow an exact, auditable path.
- **Actions and tools** through more than a thousand connectors, plus Power Automate flows and custom APIs.
- **Autonomous triggers** so an agent can start work by itself when an event happens, such as a new record or an incoming email.
- **Multi-agent orchestration** so specialised agents can call one another.
- **Multi-channel publishing** to Teams, Microsoft 365 Copilot, websites, SharePoint, voice and contact-centre platforms.
- **Analytics, evaluation and monitoring** built into the authoring surface.
- **Governance** through environments, data policies, managed environments and Microsoft Agent 365.

1.6 AI Agent Lifecycle

Every successful agent project moves through the same five stages. Day 1 of this course takes you through all five with a simple agent; the rest of the series goes deeper into each.

Stage	What you do	You are done when...
1. Create	Decide the purpose and the audience. Create the agent from the Copilot Studio home page, choosing a template, a description, or a blank agent.	The agent exists in the right environment with a clear name and description.
2. Build	Write the instructions. Add knowledge. Add tools if the agent must act. Set the scope and the limits.	The agent knows who it is, what it knows, what it can do, and where it must stop.
3. Test	Chat with it in the test canvas (Preview). Try the awkward questions, not just the easy ones. Build test sets on the Evaluate tab.	It answers the top ten real user questions correctly and refuses politely when it should.
4. Publish	Publish the agent, then connect it to the channels where your users work. Get admin approval where required.	A real user, on a real device, gets a correct answer.
5. Monitor and improve	Watch analytics and the Monitor tab. Look at unresolved conversations, escalation rates and cost. Feed what you learn back into instructions and knowledge.	You have a rhythm — weekly review, monthly improvement — rather than a one-off launch.

[SCREENSHOT 1.5]

Lifecycle diagram: Create → Build → Test → Publish → Monitor, with an arrow looping from Monitor back to Build.

BEST PRACTICE

Treat stage 5 as the real project. Most agents fail not because they were built badly, but because nobody looked at them after launch.

Use separate environments for development, test and production from the very first project. Retro-fitting this later is painful.

Publish to yourself first, test the published version, and only then widen the audience.

Step-by-step Activity 1 — Guided tour (10 minutes)

Do this with your trainer before you start Lab 1.

1. Open a browser and go to copilotstudio.microsoft.com. Sign in with your work account.
2. Look at the top right of the screen. Note the environment name shown in the environment switcher. Write it here: ______________________
3. On the home page, find the box that invites you to describe the agent you want to build. Do not type anything yet.
4. Scroll down and look at the agent templates Microsoft provides. Pick one that is closest to your own job and read its description.
5. In the left navigation, open Agents to see any agents that already exist in this environment.
6. If your tenant shows a banner or a toggle for the new experience, note which experience you are currently in, using the Tip in section 1.2.

Best Practices

- Name agents so a stranger can tell what they do: 'HR-Leave-Policy-Assistant', not 'Test1'.
- Write the agent's purpose in one sentence before you touch the product. If you cannot state it in one sentence, the scope is not clear enough yet.
- Decide who the users are and which channel they live in before you build — channel choice changes design.
- Check licensing and capacity during design, not after the pilot succeeds.
- Start narrow. One well-answered question beats twenty half-answered ones.

Common Mistakes

Mistake	Why it hurts	Do this instead
Building in the default environment	It is shared by everyone in the tenant, everyone gets the Environment Maker role automatically, and it is hard to govern.	Ask your admin for a dedicated development environment with Dataverse.
Treating an agent as a search engine	Users expect answers, not links. An agent with no instructions and ten websites attached gives vague, unciteable answers.	Narrow the knowledge, then write instructions that tell it how to answer.
Ignoring the classic-versus-new choice	Agents cannot be moved between experiences. Choosing wrongly means rebuilding.	Decide the experience deliberately, using the comparison in section 1.2.
Assuming the trial licence is enough	The trial cannot publish. Teams discover this on demo day.	Confirm licensing before the pilot starts.
Skipping the estimate of usage	Copilot Credits are consumed per response and per action. Cost surprises kill projects.	Use the agent usage estimator during design.

Knowledge Check

Answer these on your own first, then compare with the answer key in Appendix C.

Q1. Which statement best describes the difference between a copilot and an agent?

A. They are two words for the same thing.

B. A copilot assists a person who is driving the work; an agent is given a goal and decides its own steps, and can be triggered by an event.

C. A copilot can act on systems, an agent can only chat.

D. An agent must always be started by a human.

Q2. Your tenant has no Copilot Studio licence at all, and you want to try building an agent for yourself today. Which option fits?

A. Copilot Studio prepaid capacity pack

B. Copilot Credit Pre-Purchase Plan

C. Copilot Studio trial licence — but you will not be able to publish

D. Environment Maker security role on its own

Q3. Since September 2025, what is the common currency used to charge for Copilot Studio usage?

A. Messages

B. Copilot Credits

C. Power Platform requests

D. Dataverse capacity units

Q4. In the new agent experience, which four tabs make up the agent authoring surface?

A. Topics, Knowledge, Actions, Settings

B. Build, Preview, Evaluate, Monitor

C. Design, Train, Publish, Analyse

D. Create, Test, Deploy, Govern

Q5. Which statement about the classic and new agent experiences is TRUE?

A. You can convert an agent from the classic experience to the new experience at any time.

B. The new experience lets you choose between classic and generative orchestration.

C. Agents cannot be transferred between the two experiences; you choose when you create the agent.

D. The classic experience has been retired.

Key Takeaways

- Copilot Studio is a low-code, browser-based tool for building, testing, publishing and governing AI agents.
- The product grew from Power Virtual Agents; the vocabulary moved from bots and topics to agents, instructions and orchestration.
- You need access (a licence or the authors role) and capacity (Copilot Credits). They are two separate decisions.

- An agent is instructions + knowledge + tools + model, running inside a Power Platform environment and stored in Dataverse.
- The lifecycle is Create, Build, Test, Publish, Monitor — and the loop back from Monitor to Build is where value is created.

Learn More — Microsoft Learn and Official Documentation

Microsoft Learn training modules

- **Get started with Microsoft Copilot Studio (learning path) —** https://learn.microsoft.com/en-us/training/paths/work-power-virtual-agents/
- **Create agents in Microsoft Copilot Studio (learning path) —** https://learn.microsoft.com/en-us/training/paths/create-extend-custom-copilots-microsoft-copilot-studio/
- **Build an initial agent with Microsoft Copilot Studio (module) —** https://learn.microsoft.com/en-us/training/modules/create-copilots-copilot-studio/
- **Copilot Studio Agent Academy (video series) —** https://learn.microsoft.com/en-us/shows/copilot-studio-agent-academy/

Official documentation

- **Microsoft Copilot Studio documentation home —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/
- **What is Copilot Studio? —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/fundamentals-what-is-copilot-studio
- **Agents overview — new experience —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/agents-experience/overview
- **Classic vs. new agent experience —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/agents-experience/classic-vs-new
- **Copilot Studio licensing —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/billing-licensing
- **Assign licences and manage access to Copilot Studio —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/requirements-licensing
- **Agent usage estimator —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/agent-usage-estimator
- **What's new in Copilot Studio —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/whats-new

Module 2 — Environment Setup

Duration: approximately 60 minutes. Related lab: Lab 3.

Learning Objectives

By the end of this module you will be able to:

- Explain what a Power Platform environment is and why every agent needs one.
- Describe the different environment types and choose the right one for development, test and production.
- Explain what Microsoft Dataverse stores on behalf of your agent.
- Assign the security roles a maker needs, and describe the roles used for sharing and analytics.
- Find and change the most important agent settings in Copilot Studio.
- Describe the publishing options available and what each one requires.

2.1 Power Platform Environment

An environment is a container. It is a space that stores, manages and shares your organisation's business data, apps, flows and agents. Every agent you create lives in exactly one environment. Environments can have different security settings, different data, different regions and different audiences.

[SCREENSHOT 2.1]

The Power Platform admin centre, Environments list, showing name, type, region and Dataverse status.

Why organisations use more than one environment

- To separate work in progress from what real users see — a development environment, a test environment and a production environment.
- To give each department or business unit its own space with its own data.
- To meet data residency requirements by creating environments in specific geographic regions.
- To keep different global branches of the company apart.

Environment type	What it is for	Notes for Copilot Studio
Default	Created automatically for every tenant. Everyone in the tenant can use it.	Avoid building real agents here. Every user is automatically an Environment Maker, and that role cannot be removed, which makes governance very hard.
Developer	A personal, free environment for one maker to experiment in.	Excellent for personal learning. Not for shared or production work.
Sandbox	A non-production environment for building and testing. Can be reset and copied.	The normal home for a development or test agent.
Production	For live use by real users, backed by support commitments.	Where the published agent lives. Restrict who can create agents here.
Trial	Time-limited environment for evaluation.	Expires. Do not build anything you want to keep.

WATCH OUT

An agent needs a Dataverse database in its environment. If you cannot see Copilot Studio content in an environment, the first thing to check is whether Dataverse has been provisioned there.

Creating an environment (administrator task)

1. Sign in to the Power Platform admin centre at admin.powerplatform.microsoft.com with an administrator account.
2. In the left navigation, select Manage, then Environments.
3. Select + New.

4. Enter a clear name — for example 'Contoso — Copilot Studio — DEV'. Choose the region, and choose the environment type (Sandbox for development).
5. Turn on the option to create a Dataverse database for this environment.
6. Specify the Dataverse details: language, currency, and optionally a security group that limits who can access the environment. Select Save.
7. Select Refresh on the Environments list. It can take a few minutes before the new environment appears as ready.
8. Return to Copilot Studio and pick the new environment from the environment switcher in the top command bar.

TIP

Attach a Microsoft Entra security group to the environment when you create it. Then you control who can see the environment simply by managing group membership — no per-user work later.

BEST PRACTICE

Use a naming convention that includes the company, the purpose and the stage: '<Org> — <Workload> — DEV / TEST / PROD'.

Create the environment in the region closest to your users, for both latency and data residency.

Consider promoting Copilot Studio environments to managed environments — this unlocks extra administrative controls and pipelines.

Keep Copilot Studio agents in dedicated environments rather than mixing them with canvas apps, so licensing and governance stay simple.

2.2 Security Roles

Having a licence does not, by itself, let you build an agent in a particular environment. You also need a security role inside that environment. Security roles are a Dataverse concept: they control which tables a user can read, create, update and delete, and at what scope.

Role	Who gets it	What it allows
Environment Maker	Makers who need to author agents.	Create and edit agents in the environment. This is the standard maker role. It is also required for anyone you share an agent with as a co-author.
System Administrator	Environment administrators only.	Full control of the environment, including assigning roles to others. Needed if you must grant Environment Maker to a co-author who does not have it yet.
System Customizer	Advanced makers who customise Dataverse.	Create and customise tables, columns and solutions without full admin rights.
Basic User	End users who only consume.	Run apps and interact with data they own. Suitable for most users in a production environment.
Analytics Viewer	Analysts, product owners, support leads.	Access to a specific agent's Analytics page and its metrics, and the ability to open the agent from there.
Bot Transcript Viewer	Support and quality teams.	Read session-level conversation transcripts stored in Dataverse. Assigned at environment level in the Power Platform admin centre.

Assigning a security role (administrator task)

1. In the Power Platform admin centre, select Manage, then Environments, and open your environment.
2. Under Access, select See all next to Users. (You can also use Settings, then Users + permissions, then Users.)
3. Search for the user. If they do not appear, they may not have been added to the environment yet.
4. Select the user, then select Manage security roles.
5. Tick Environment Maker (or the role you need) and select Save.

The better way — group teams

Assigning roles to individuals does not scale. In real projects, roles are assigned to teams that are backed by Microsoft Entra ID groups.

1. In the Power Platform admin centre, open the environment and go to Settings, then Users + permissions, then Teams.
2. Select + Create team. Give it a meaningful name, such as 'Copilot Studio Makers — DEV'.
3. For Team type, choose Microsoft Entra ID Security Group (or Office Group).
4. Search for and select the Entra group, choose the membership type, then select Next.
5. In Manage security roles, select the role — for example Environment Maker — and select Save.
6. From now on, adding a person to the Entra group gives them the role automatically.

BEST PRACTICE

Assign licences through Entra ID groups, not to individuals.
Assign security roles through group teams inside each Dataverse environment.
Give production users a restrictive role such as Basic User; keep Environment Maker for the development and test environments.
Apply data policies (DLP) so agents can only use the connectors your project actually needs.
Review who has access to each environment on a schedule — access granted for a pilot has a habit of surviving for years.

WATCH OUT

In the default environment, every user in the tenant automatically holds the Environment Maker role, and it cannot be removed. This is the strongest practical reason to build somewhere else.

2.3 Dataverse Basics

Microsoft Dataverse is the secure data platform underneath Power Platform. For Copilot Studio it plays two roles: it is where your agent itself is stored, and it can be a knowledge source or an action target for the agent.

Dataverse term	In plain language
Table	A collection of records of the same kind — like a spreadsheet with a strict structure. Formerly called an entity.
Column	One field on a table, with a data type. Formerly called an attribute.
Row / record	One item in a table — one customer, one ticket, one agent definition.
Relationship	A defined link between two tables, such as one account having many contacts.
Solution	A container used to package and move components between environments.
Security role	A set of permissions on tables, at a given scope (own records, business unit, whole organisation).
Business unit	An organisational grouping used to scope who can see which records.

What Copilot Studio stores in Dataverse

- The agent definition itself, in the bots table, and its sub-components in botcomponents — topics, entities and other building blocks.
- Conversation transcripts, which are what analytics and troubleshooting are built on.
- Sharing information: when you create an agent, a team is created and the agent and its components are shared with that team.
- Solution metadata, so agents can be packaged and moved from development to test to production.

> **NOTE**
> Conversation transcripts are shared implicitly with the parent agent's team, but only users who have read access to the Conversation Transcript table can actually open them. This is why the Bot Transcript Viewer role exists.

[SCREENSHOT 2.2]

Dataverse tables view showing the bots and botcomponents tables in a Copilot Studio environment.

2.4 Copilot Settings

Once your agent exists, a range of settings control how it behaves. Where these live depends on which experience you are in — in the classic experience, Settings is an item in the left navigation; in the new experience, most of it sits on the Build tab and in the agent's details panel.

Setting area	What you control	Typical Day 1 choice
Details	Agent name, description, icon, and the language the agent uses.	A clear name and a one-line description of the purpose.
Instructions	The agent's persona, tone, scope and rules.	See Module 3, section 3.2.
Knowledge	Which sources the agent may read, and how it uses them.	Start with one trusted source.
Generative AI / orchestration	In the classic experience, whether the agent uses classic or generative orchestration, and whether it may use general knowledge. In the new experience, orchestration is fixed and there is no general-knowledge toggle.	For a grounded internal agent, restrict answers to your own sources.
Security	Authentication for the agent: no authentication, authenticate with Microsoft, or manual authentication. Also web channel security.	Authenticate with Microsoft for an internal agent.

Setting area	What you control	Typical Day 1 choice
Channels	Where the agent is available — Teams and Microsoft 365 Copilot, websites, SharePoint, and more.	Teams and Microsoft 365 Copilot for an internal agent.
Entities and variables	Reusable data definitions and values carried across a conversation.	Covered on Day 2.
Analytics	Usage, resolution rate, escalation rate, customer satisfaction and session data.	Review after every pilot week.

[SCREENSHOT 2.3]

The agent Settings page in the classic experience, with the Details, Security and Generative AI sections visible.

WATCH OUT

Turning off authentication makes an agent available to anonymous users. Never do this for an agent that can reach internal data. Decide the authentication setting before you add any knowledge source.

2.5 Publishing Options

Publishing takes the version of the agent you have been editing and makes it the live version. Nothing you change reaches your users until you publish again.

Step	What happens
1. Publish	You select Publish in the agent. Copilot Studio validates the agent and creates the live version. Publishing applies to every channel the agent is connected to.
2. Connect a channel	After the first publish, you connect the agent to one or more channels — Microsoft 365 Copilot and Teams, a website, SharePoint, Power Pages, a voice channel, a contact centre, or a custom app through Direct Line.
3. Set availability	For Teams and Microsoft 365 Copilot you decide whether to install for yourself, share a link with specific people, or submit the agent for admin approval so it appears in the organisation's agent store.
4. Re-publish after changes	Every subsequent change needs another publish. Users who are already in a persistent conversation may keep the old version until the session resets.

Channel family	Examples	Best for
Microsoft experiences	Microsoft 365 Copilot, Microsoft Teams, SharePoint, Power Pages	Internal employee agents. Users are already signed in and identity flows through.
Websites and apps	Custom website embed, custom client via Direct Line API	Customer-facing agents on your own web properties.
Voice and telephony	Voice channel, contact-centre integrations	Call deflection and support scenarios.
Messaging and contact centre	Messaging platforms, Dynamics 365 Contact Center and other engagement hubs	Escalation to human agents and omnichannel support.

NOTE

Publishing an agent to Microsoft 365 Copilot and Teams is generally available. Unlike simple declarative agents, a Copilot Studio agent is not distributed automatically when you publish — you must set its availability, and organisation-wide publishing normally needs administrator approval.

TIP

In a persistent channel such as Teams, type 'start over' in the chat to reset the session and pick up your newly published version immediately, instead of waiting for the refresh delay.

Step-by-step Activity 2 — Check your own access (10 minutes)

1. Open Copilot Studio and note the environment shown in the environment switcher.

2. If you have administrator rights, open the Power Platform admin centre and confirm that the environment has a Dataverse database.
3. Find your own user record in the environment and list the security roles you hold. Write them here: ____________________
4. If you do not hold Environment Maker, note down what you would ask your administrator for, in one sentence.
5. Open your agent's Settings and note the current authentication option.

Best Practices

- Three environments as a minimum: development, test and production. Never demo from production.
- Manage access through Entra ID groups and Dataverse group teams, never user by user.
- Provision Dataverse when you create the environment, not as an afterthought.
- Apply data loss prevention policies to block connectors and channels your project does not need.
- Document the environment strategy on one page and share it with every maker on day one.

Common Mistakes

Mistake	Why it hurts	Do this instead
Building in the default environment	Uncontrolled access, no clean way to promote to production.	Use a dedicated sandbox environment.
Giving everyone System Administrator	Removes every safety net and fails audit.	Environment Maker for makers, Basic User for consumers.
Forgetting Dataverse	Agents cannot be created or stored.	Provision Dataverse at environment creation.
Editing directly in production	No rollback, and users see half-finished work.	Build in development, move with solutions or pipelines.
Assuming an edit is live	Users keep getting the old answers and trust drops.	Publish after every change, and confirm in the channel.

Knowledge Check

Answer these on your own first, then compare with the answer key in Appendix C.

Q1. What must an environment have before you can create a Copilot Studio agent in it?

A. A Power BI workspace
B. A Dataverse database
C. A SharePoint site collection
D. An Azure subscription

Q2. Which security role is the standard role for a maker who needs to author agents?

A. Basic User
B. System Administrator
C. Environment Maker
D. Analytics Viewer

Q3. A colleague needs to view an agent's Analytics page but must not be able to edit the agent. Which role fits best?

A. Environment Maker
B. Analytics Viewer
C. System Customizer
D. Bot Transcript Viewer

Q4. Why is the default environment a poor choice for building production agents?

A. It does not support Dataverse.
B. Every user in the tenant automatically holds the Environment Maker role there and it cannot be removed.
C. Agents built there cannot be published.
D. It is always in a different region.

Q5. You have changed your agent's instructions. Users in Teams still get the old behaviour. What is the most likely reason?

A. The agent needs a new licence.
B. You have not published the agent since making the change.
C. Dataverse has not synchronised.
D. The Teams channel must be deleted and recreated.

Key Takeaways

- An environment is the container for your agent, and Dataverse inside it is where the agent actually lives.
- Access requires both a licence and a security role in the target environment. Environment Maker is the maker's role.
- Manage licences and roles through Entra ID groups and Dataverse group teams so access scales and can be audited.
- Publishing creates the live version; connecting channels decides where users meet it. Both steps are needed.
- Every change needs a new publish before users see it.

Learn More — Microsoft Learn and Official Documentation

Microsoft Learn training modules

- **Get started with Microsoft Dataverse (learning path) —** https://learn.microsoft.com/en-us/training/paths/get-started-with-powerapps-common-data-service/
- **Get started with security roles in Dataverse (module) —** https://learn.microsoft.com/en-us/training/modules/get-started-security-roles/
- **Administer Microsoft Power Platform (learning path) —** https://learn.microsoft.com/en-us/training/paths/administrate-power-platform/

Official documentation

- **Work with Power Platform environments in Copilot Studio —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/environments-first-run-experience
- **Environments overview (Power Platform) —** https://learn.microsoft.com/en-us/power-platform/admin/environments-overview
- **Create an environment —** https://learn.microsoft.com/en-us/power-platform/admin/create-environment
- **Security concepts in Microsoft Dataverse —** https://learn.microsoft.com/en-us/power-platform/admin/wp-security-cds
- **Configure user security in an environment —** https://learn.microsoft.com/en-us/power-platform/admin/database-security
- **Share agents with other users —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/admin-share-bots
- **Security and governance in Copilot Studio —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-and-governance
- **Secure your Copilot Studio projects —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/guidance/sec-gov-phase3
- **Key concepts — publish and deploy your agent —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/publication-fundamentals-publish-channels
- **Available channels for agents (new experience) —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/agents-experience/publication-channels-overview
- **Data loss prevention policies —** https://learn.microsoft.com/en-us/power-platform/admin/wp-data-loss-prevention

Module 3 — Build Your First Copilot

Duration: approximately 45 minutes of concept, then Labs 4, 5 and 6.

Learning Objectives

By the end of this module you will be able to:

- Create an agent in Copilot Studio using a description, a template or a blank start.
- Write clear instructions that set the agent's identity, scope, tone and limits.
- Add and configure knowledge sources so the agent answers from your content instead of guessing.
- Test the agent in the test canvas, read the citations, and interpret what the agent did.
- Recognise when an answer is wrong because of instructions, because of knowledge, or because of scope.

3.1 Creating a Copilot

There are three ways to start. All three produce the same kind of agent — they just differ in how much you write yourself.

Starting point	How it works	Use it when
Describe it	You type a plain-language description of what you want. Copilot Studio drafts the agent's name, description and instructions for you, and you refine them by conversation.	You know the outcome but not the product. This is the fastest way to a working draft.
Template	You pick from Microsoft's library of ready-made agents — for example a safe travels agent, an IT helpdesk agent or a website Q&A agent — and change what you need.	Your scenario is close to a common pattern.
Blank / skip to configure	You start with an empty agent and configure every part yourself.	You already know exactly what you want, or you are following a governed internal standard.

Creating an agent — the steps

1. Go to copilotstudio.microsoft.com and sign in with your work account.
2. Check the environment switcher in the top command bar. Make sure you are in your development environment, not the default one.
3. Select Create in the left navigation, or use the description box on the home page.
4. Choose New agent. If your tenant offers both experiences, choose the experience you want — remember, an agent cannot be moved between them later.
5. Describe the agent in one or two sentences. For example: 'An assistant for Contoso employees that answers questions about the leave policy using the HR policy documents, in a friendly and concise tone.'
6. Review the name, description and instructions that are generated. Edit them so they say exactly what you mean.
7. Select Create. The agent opens on its Build tab (new experience) or overview page (classic experience).

[SCREENSHOT 3.1]

The Create agent screen with the natural-language description box and the option to skip to configure.

NOTE

Creating your first agent in an environment also installs the Copilot Studio solutions into that environment. The first creation can therefore take a little longer than later ones.

TIP

Give the agent a name your users will see and recognise in Teams — 'HR Assist', not 'CS-Agent-POC-v3'. Keep the internal version number in the description instead.

3.2 Configure Instructions

Instructions are the single most important thing you write. They define the agent's identity, tone, scope and behaviour. In the new agent experience, instructions and reasoning replace the explicit topic flows of the classic experience — the agent decides what to do by reading what you wrote.

A simple structure that works

Part	What to write	Example
Role	Who the agent is and who it serves.	"You are the HR assistant for Contoso India employees."
Scope	What it should answer and, just as important, what it should not.	"Answer only questions about leave, attendance and holidays. Do not answer questions about payroll or performance reviews."
Sources	Where answers must come from.	"Use only the HR policy documents provided as knowledge. Do not use general knowledge."
Tone and format	How it should sound and how answers should look.	"Be warm, brief and practical. Use short bullet points. Always give the exact policy clause number."
Fallback	What to do when it does not know.	"If the answer is not in the documents, say you are not sure and give the HR helpdesk email hr@contoso.com."
Safety and privacy	Any hard limits.	"Never ask for or repeat personal identification numbers. Never give legal advice."

[SCREENSHOT 3.2]

The instructions box on the Build tab, with a multi-paragraph instruction written for an HR assistant.

BEST PRACTICE

Write instructions as if you were briefing a competent new joiner on their first day — clear, specific, and explicit about the boundaries.

Be positive and concrete. 'Answer in three bullet points or fewer' works far better than 'do not be verbose'.

Always include a fallback sentence. An agent that admits it does not know earns more trust than one that improvises.

Change one thing at a time and re-test. If you change five lines at once, you will not know which one helped.

Keep a copy of the instruction text outside the product, so you can diff and roll back.

WATCH OUT

Very long instructions are not automatically better. Contradictory or rambling instructions make the agent inconsistent. Aim for clear and structured rather than exhaustive.

3.3 Knowledge Sources

Knowledge is what the agent is allowed to read when it answers. Adding knowledge is what turns a general-purpose assistant into your organisation's assistant. Answers grounded in your own content come back with citations, so users can check the source.

Knowledge source	What it is good for	Things to check
Uploaded files	Policies, manuals, product sheets, FAQs. Drag and drop directly into the Add knowledge dialog.	File type and size limits. Keep documents current — the agent will happily quote last year's policy.
Public websites	Published product documentation, public FAQ pages, help centres.	Only public, indexable pages. Point at specific sections rather than an entire domain.

Knowledge source	What it is good for	Things to check
SharePoint	Internal document libraries and intranet content.	The agent respects the user's own permissions, so different users may see different answers. That is correct behaviour.
Dataverse	Structured business records held in Power Platform.	Choose the specific tables. Do not point at everything.
Azure AI Search	Large or specialised indexes you already maintain.	Requires an existing index and connection details.
Dynamics 365, Salesforce, ServiceNow, Azure SQL	Line-of-business records — cases, orders, incidents, customers.	Needs a connection and appropriate credentials.
Microsoft IQ (new experience)	Organisational data from Microsoft 365 — email, calendar events, files, Teams messages and people information.	Powerful and personal. Confirm your governance position before you enable it.

Adding a knowledge source — the steps (new experience)

1. Open your agent and select the Build tab.
2. In the components panel on the right, select Knowledge.
3. In the Add knowledge dialog, either drag and drop your file onto the upload area, search for a source, or pick a source type such as Upload file, Public websites, SharePoint, Azure AI Search, Dataverse, Dynamics 365, Salesforce, ServiceNow or Azure SQL.
4. Complete the configuration for the source you chose — a URL, a site address, a connection, or a table selection.
5. Select Add or Save.
6. Wait for the source to finish processing. Large documents and websites take a few minutes before answers improve.
7. Open the Preview tab and ask a question you know the answer to. Check that the citation points at the right document.

[SCREENSHOT 3.3]

The Add knowledge dialog showing the drag-and-drop upload area and the list of available source types.

NOTE

In the new agent experience there is no general-knowledge on/off toggle. What the agent can draw on is controlled by the specific sources you add. Microsoft IQ is configured separately from the components panel.

BEST PRACTICE

Start with one authoritative source. Add a second only after the first is answering well.
Prefer a curated, well-structured document over a whole website. Quality of source beats quantity every time.
Name each source clearly and add a description so future makers know why it is there.
Agree an owner and a refresh cycle for every source before go-live — stale knowledge is the most common cause of wrong answers in month three.
Check citations during testing, not just the wording of the answer.

3.4 Testing the Agent

The test canvas is the chat panel you use while you build. In the classic experience it is the test pane beside the authoring canvas; in the new experience it is the Preview tab. It uses the version you are editing, not the published one, so you can try changes safely.

What to test	Why	Example prompt
A question you know the answer to	Confirms grounding and citations work.	"How many casual leave days do I get in a year?"
The same question asked differently	Real users do not use your words.	"cl balance policy?"
A question that is in scope but not in the documents	Confirms the fallback behaves.	"What is the policy for sabbatical leave?"
A question that is clearly out of scope	Confirms the boundary holds.	"What is my salary this month?"
An ambiguous question	Checks whether the agent asks a clarifying question instead of guessing.	"Can I take leave next week?"
A rude or nonsense message	Checks tone and resilience.	"asdkjh" or an unfriendly message.

[SCREENSHOT 3.4]

The Preview / test canvas with a conversation in progress and a citation link shown under the answer.

Reading what the agent did

- **Citations** under an answer tell you which source was used. No citation usually means the answer was not grounded.
- **The activity or trace view** shows which knowledge source, topic or tool the agent chose. In the new experience the Monitor tab shows recent tasks and what the agent accessed.
- **The Evaluate tab** (new experience) lets you build a test set and run it repeatedly, so you can measure quality instead of relying on a few lucky prompts.

Symptom	Most likely cause	First thing to try
Answers are vague and have no citation	The agent is not using your knowledge, or the source is still processing.	Confirm the source status; ask a question using words that appear in the document.
Answers are correct but too long or off-tone	Instructions do not say how to answer.	Add explicit tone and format rules to the instructions.
The agent answers questions it should refuse	Scope is not stated in the instructions.	Add an explicit 'do not answer' list and a fallback sentence.
Different users get different answers from SharePoint	Working as designed — SharePoint respects each user's permissions.	Verify with a test user; document the behaviour for your stakeholders.
Changes do not appear for users	The agent has not been published since the change.	Publish, then reset the session in the channel.

Best Practices

- Build the smallest agent that solves one real problem, then grow it.
- Write instructions first, add knowledge second, add tools last.
- Test with the questions real users actually ask — collect twenty of them before you start building.
- Keep a simple test script and re-run it after every significant change.
- Record who owns the agent, who owns each knowledge source, and how often each is refreshed.

Common Mistakes

Mistake	Why it hurts	Do this instead
Adding ten knowledge sources on day one	The agent cannot tell which source is authoritative and answers become inconsistent.	One source, tested, then add more.

Mistake	Why it hurts	Do this instead
Leaving the generated instructions untouched	Generated text is a starting point, not a specification.	Rewrite it in your own words using the six-part structure.
Testing only the happy path	Users find the edges within an hour of launch.	Test out-of-scope, ambiguous and hostile prompts too.
Confusing the test canvas with production	The test canvas uses your draft; users see the published version.	Always confirm behaviour in the real channel after publishing.
Uploading confidential documents without checking governance	Knowledge sources inherit the sensitivity of the content you add.	Confirm classification and DLP policy before uploading anything.

Knowledge Check

Answer these on your own first, then compare with the answer key in Appendix C.

Q1. In the new agent experience, from which tab do you add knowledge, tools and instructions?

A. Preview
B. Build
C. Monitor
D. Evaluate

Q2. Which part of an instruction tells the agent what to do when it cannot find an answer?

A. Role
B. Scope
C. Fallback
D. Tone

Q3. Your agent gives a plausible answer but there is no citation underneath it. What is the most useful first check?

A. Republish the agent.
B. Check whether the knowledge source has finished processing and is actually attached.
C. Change the model.
D. Add another website as a knowledge source.

Q4. Two employees ask the same question and get different answers from a SharePoint knowledge source. What is happening?

A. The agent is broken and must be recreated.
B. SharePoint knowledge respects each user's own permissions, so they see different content.
C. The publish did not complete.
D. The instructions contain a contradiction.

Q5. Which sequence reflects good practice when building a first agent?

A. Add ten knowledge sources, then write instructions, then test.
B. Write instructions, add one knowledge source, test, then extend.
C. Publish first, then configure, then test.
D. Add tools first, then knowledge, then instructions.

Key Takeaways

- You can create an agent from a description, from a template, or from blank — the description route is the fastest way to a working draft.
- Instructions define identity, scope, sources, tone and fallback. They are the highest-leverage thing you write.
- Knowledge grounds the agent in your content and produces citations. Start with one authoritative source.
- The test canvas uses your draft version. Test the edges, not just the happy path, and read the citations.
- Most quality problems trace back to one of three things: unclear instructions, weak knowledge, or undefined scope.

Learn More — Microsoft Learn and Official Documentation

Microsoft Learn training modules

- **Create agents with Microsoft Copilot Studio — Online Workshop (learning path) —** https://learn.microsoft.com/en-us/training/paths/power-virtual-agents-workshop/
- **Build an initial agent with Microsoft Copilot Studio (module) —** https://learn.microsoft.com/en-us/training/modules/create-copilots-copilot-studio/
- **Agent in a Day — Online Workshop (learning path) —** https://learn.microsoft.com/en-us/training/paths/agents-online-workshop/
- **Enhance agents with autonomous capabilities (learning path) —** https://learn.microsoft.com/en-us/training/paths/enhance-autonomous-agents/

Official documentation

- **Create and delete agents —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/authoring-first-bot
- **Configure agent details and instructions (new experience) —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/agents-experience/authoring-instructions
- **Add knowledge sources to an agent (new experience) —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/agents-experience/knowledge-add-existing-copilot
- **Knowledge sources summary —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-copilot-studio
- **Use Microsoft IQ (new experience) —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/agents-experience/use-microsoft-iq
- **Add tools and skills to an agent —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/agents-experience/tools-skills-legacy
- **Select a model for an agent —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/agents-experience/authoring-select-agent-model
- **Test your agent —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/authoring-test-bot
- **AI-based agent authoring overview —** https://learn.microsoft.com/en-us/microsoft-copilot-studio/nlu-gpt-overview

Hands-on Labs — Day 1

The six labs below build on each other. Complete them in order using the same agent throughout, unless your trainer says otherwise. Write your answers and observations directly in this handbook — you will need them on Day 2.

Lab	Title	You will produce
1	Create Your First Agent	A working agent in your development environment.
2	Explore Copilot Studio Interface	A completed interface map and vocabulary sheet.
3	Configure Environment and Security Roles	A documented environment with roles assigned correctly.
4	Customize Agent Instructions	Instructions rewritten in the six-part structure, plus one knowledge source.
5	Test Conversations Using Test Canvas	A completed six-prompt test script with results.
6	Publish the First Version of the Copilot	A published agent available to you in Teams or Microsoft 365 Copilot.

> **WATCH OUT**
> Before you start: confirm with your trainer which environment you should be using and which experience — classic or new — your classroom tenant has. Where the two experiences differ, the steps below call it out.

Lab 1 — Create Your First Agent

Duration	Objective	Prerequisites
20 minutes	Create a working agent from a natural-language description and confirm it exists in the correct environment.	Copilot Studio access; Environment Maker role in a Dataverse-enabled environment.

Scenario

Contoso employees keep emailing HR the same questions about leave. You will build an assistant that answers those questions from the HR policy documents.

Steps

1. Open a browser and go to copilotstudio.microsoft.com. Sign in with your work account.
2. Look at the environment switcher in the top command bar. If it does not show your development environment, select it and change it now.
3. On the home page, find the description box. Type: 'An assistant for Contoso employees that answers questions about the leave policy. It should be friendly, brief, and always cite the policy document. If it does not know, it should give the HR helpdesk email.'
4. Press Enter or select the send arrow. Copilot Studio drafts a name, description and instructions.
5. Read what it generated. Change the name to 'HR Leave Assistant — <your initials>'.
6. If you are offered a choice of experience, note which one you selected and write it here: ____________________
7. Select Create. Wait for the agent to open. The first agent in a new environment takes longer because solutions are being installed.
8. Take a moment to look at the top of the screen and identify the Publish button and the settings entry point.

[SCREENSHOT L1.1]

The newly created agent open on its Build tab (new experience) or overview page (classic experience).

CHECKPOINT — YOU ARE DONE WHEN

The agent appears under Agents in your development environment.
Its name identifies both the purpose and you.
You can open it and see its generated description and instructions.

If something goes wrong

Symptom	What to do
'Create' is greyed out or you get a permissions error	You are missing a licence or the Environment Maker role. Check with your trainer or administrator, and see Module 1 section 1.3 and Module 2 section 2.2.
No environments appear in the switcher	You have not been added to any environment, or the environment has no Dataverse database.
Creation takes several minutes	This is normal for the first agent in an environment — Copilot Studio is installing its solutions.

Reflection

- How closely did the generated instructions match what you actually meant?
- What would you change before letting a real employee use this agent?

Lab 2 — Explore Copilot Studio Interface

Duration	Objective	Prerequisites
20 minutes	Build a mental map of the authoring interface so you can find anything quickly for the rest of the course.	Lab 1 complete.

Steps

1. With your agent open, identify the environment switcher. Write the environment name here: ______________________
2. Identify how you get back to the list of all agents in this environment.
3. Find where instructions are edited. In the new experience this is on the Build tab; in the classic experience it is in the agent's overview or settings.
4. Find where knowledge sources are added. New experience: the Knowledge button in the components panel on the Build tab. Classic experience: the Knowledge item in the left navigation.
5. Find where tools or actions are added.
6. Find the test canvas. New experience: the Preview tab. Classic experience: the Test your agent pane, usually opened from the top right.
7. Find the analytics or Monitor area.
8. Find the Publish button and, without selecting it, note where the channel configuration lives.
9. Complete the interface map table below.

Your interface map

I want to...	Where I found it in this tenant
Change what the agent says about itself	____
Add a document the agent can read	____
Let the agent do something in another system	____
Chat with the agent while I build	____
See how the agent is performing	____
Make my changes live	____
Choose where users can find the agent	____
Switch to a different environment	____

[SCREENSHOT L2.1]

The agent authoring surface with the navigation elements numbered and labelled.

CHECKPOINT — YOU ARE DONE WHEN

Every row of the interface map is filled in.

You can state which experience — classic or new — you are working in, and why you can tell.

TIP

Keep this page open for the rest of the day. Trainers move fast in the later labs, and this map will keep you with them.

Lab 3 — Configure Environment and Security Roles

Duration	Objective	Prerequisites
35 minutes	Understand and, where you have rights, configure the environment and security roles a Copilot Studio project needs.	Power Platform admin centre access, or the read-only path described below.

NOTE
Not everyone in the room will have administrator rights. Path A is for those who do. Path B is a read-only investigation that reaches the same understanding. Your trainer will tell you which to follow.

Path A — with administrator rights

1. Go to admin.powerplatform.microsoft.com and sign in.
2. Select Manage, then Environments. Find your development environment and note its type and region.
3. Open the environment and confirm that a Dataverse database exists.
4. Under Access, select See all next to Users. Find your own user record.
5. Select Manage security roles and confirm that Environment Maker is assigned. If it is not, assign it and save.
6. Now create a group team: go to Settings, then Users + permissions, then Teams, and select + Create team.
7. Name it 'Copilot Studio Makers — DEV'. For Team type choose Microsoft Entra ID Security Group, select an existing group, choose the membership type and select Next.
8. In Manage security roles, select Environment Maker and save.
9. Optional, if your trainer provides a second user: share your agent with them and observe that Copilot Studio requires them to hold Environment Maker before co-authoring works.

Path B — without administrator rights

1. In Copilot Studio, note the environment name from the environment switcher.
2. Open your agent's settings and record the current authentication option.
3. Using Module 2 as your reference, write down the exact request you would send to your administrator: which environment, which role, for whom, and why. Draft it in the box below.
4. List the three roles you would assign in a production environment for: a maker, a support analyst who needs analytics, and an ordinary employee.

Your access request draft
____
____
____
____

[SCREENSHOT L3.1]

Power Platform admin centre: the Manage security roles panel for a user, with Environment Maker selected.

CHECKPOINT — YOU ARE DONE WHEN
You can name your environment, its type, its region, and confirm it has Dataverse.
You can state which security roles you hold and which ones you would need for co-authoring.
You can explain in one sentence why the default environment is a poor place to build.

If something goes wrong

Symptom	What to do
The user does not appear in the environment's user list	The user has never accessed the environment, or is blocked by the environment's security group. Add them to the group first.
You cannot create a team	You need System Administrator on the environment.
Sharing an agent fails with a permissions message	The co-author needs the Environment Maker role. You need System Administrator for Copilot Studio to assign it on their behalf.

Lab 4 — Customize Agent Instructions

Duration	Objective	Prerequisites
30 minutes	Rewrite your agent's instructions using the six-part structure, and add one knowledge source.	Labs 1 and 2 complete. One PDF or a public website URL to use as knowledge.

Part 1 — Rewrite the instructions

1. Open your agent and go to the instructions. New experience: the Build tab. Classic experience: the agent's overview or settings.
2. Read the generated instructions and copy them into the notes area at the end of this lab, so you can compare later.
3. Replace them with your own version, using the six-part structure from Module 3 section 3.2: Role, Scope, Sources, Tone and format, Fallback, Safety.
4. Use the worked example below as your model, but change it to fit your own scenario.
5. Save your changes.

Worked example — instructions for the HR Leave Assistant
You are the HR leave assistant for Contoso India employees.
Answer only questions about leave, attendance, public holidays and the leave approval process. Do not answer questions about payroll, appraisals, recruitment or anything outside HR.
Use only the HR policy documents provided to you as knowledge. Do not rely on general knowledge, and do not invent policy.
Be warm, brief and practical. Answer in three short bullet points or fewer. Always name the policy document and clause you used.
If the answer is not in the documents, say clearly that you are not sure, and give the HR helpdesk address hr@contoso.com.
Never request, repeat or store personal identification numbers. Never give legal or tax advice.

Part 2 — Add a knowledge source

1. New experience: on the Build tab, select Knowledge in the components panel. Classic experience: select Knowledge in the left navigation, then Add knowledge.
2. Choose Upload file and drag in your PDF, or choose Public websites and enter your URL.
3. Give the source a clear name and a one-line description of what it contains.
4. Select Add or Save, and wait for the source to finish processing.
5. Open the test canvas and ask one question you know the document answers. Confirm the citation points at your source.

[SCREENSHOT L4.1]

Instructions rewritten in the six-part structure, with one knowledge source listed in the components panel.

CHECKPOINT — YOU ARE DONE WHEN

Your instructions contain all six parts and are written in your own words.
One knowledge source is attached, named and processed.
A test question returns an answer with a citation to your source.

If something goes wrong

Symptom	What to do
The agent still answers in long paragraphs	Your tone and format rule is not explicit enough. Say the maximum number of bullet points.

Symptom	What to do
The agent answers out-of-scope questions	Add an explicit list of topics it must refuse, and re-test that exact question.
The website source returns nothing useful	The page may not be publicly indexable, or it may be too broad. Try a specific page rather than a domain root.
Answers have no citation	The source may still be processing, or your question uses words that do not appear in the document.

Notes — original generated instructions

Lab 5 — Test Conversations Using Test Canvas

Duration	Objective	Prerequisites
30 minutes	Run a structured test script against your agent and record what happened, instead of testing at random.	Lab 4 complete.

Steps

1. Open the test canvas. New experience: the Preview tab. Classic experience: the Test your agent pane.
2. Run each of the six prompts in the table below, one at a time.
3. For each one, record the answer quality, whether there was a citation, and anything surprising.
4. After the sixth prompt, choose the single weakest result and improve the agent — change one thing only, in instructions or knowledge.
5. Re-run that one prompt and record whether it improved.
6. If your tenant has the Evaluate tab, create a test set from these prompts and run it.

Your test script

#	Prompt type	What you typed	Answer OK?	Citation?	Notes
1	Known answer	______________	Y / N	Y / N	___________
2	Same question, different words	______________	Y / N	Y / N	___________
3	In scope, not in the documents	______________	Y / N	Y / N	___________
4	Clearly out of scope	______________	Y / N	Y / N	___________
5	Ambiguous question	______________	Y / N	Y / N	___________
6	Nonsense or rude message	______________	Y / N	Y / N	___________

[SCREENSHOT L5.1]

The test canvas with an answer, its citation, and the activity or trace view expanded.

The one change you made

What you changed	Prompt you re-ran	Did it improve?
______________________________	________________	Y / N

CHECKPOINT — YOU ARE DONE WHEN

All six prompts have been run and recorded.

You made exactly one change and measured its effect.

You can explain whether your weakest result was an instructions problem, a knowledge problem or a scope problem.

BEST PRACTICE

Keep this test script. Re-run it after every meaningful change to the agent — it becomes your regression test.

Change one thing at a time. Two changes at once tell you nothing about which one worked.

Lab 6 — Publish the First Version of the Copilot

Duration	Objective	Prerequisites
30 minutes	Publish your agent, connect it to a channel, and confirm that the published version behaves as expected.	Lab 5 complete. A licence that permits publishing — the trial licence does not.

WATCH OUT
If you are using a trial licence you can complete steps 1 to 3 as a walkthrough with your trainer, but the publish itself will not succeed. Note this in your reflection rather than trying to work around it.

Steps

1. Before publishing, re-read your instructions once more and fix anything you would not want a real user to see.
2. Select Publish at the top of the agent. Confirm in the dialog.
3. Wait for publishing to finish. Note how long it took: __________
4. Open the channels area. Select Teams and Microsoft 365 Copilot.
5. If you want the agent available in both, make sure the option to make the agent available in Microsoft 365 Copilot is selected. Then select Add channel.
6. Review the agent's appearance details — display name, short description, long description and icon. These are what your users see, so make them clear.
7. Choose availability: install the agent for yourself first. Do not submit it for organisation-wide approval in this lab.
8. Open Microsoft Teams or the Microsoft 365 Copilot app, find your agent, and ask it prompt 1 from your Lab 5 test script.
9. Compare the published answer with the answer you got in the test canvas. Record any difference.
10. Make one small change to the instructions in Copilot Studio and, without publishing, ask the question again in Teams. Confirm that nothing changes. Then publish and try again.

[SCREENSHOT L6.1]
The publish confirmation dialog, followed by the Teams and Microsoft 365 Copilot channel configuration panel.

CHECKPOINT — YOU ARE DONE WHEN
The agent shows as published.
The Teams and Microsoft 365 Copilot channel is connected and the agent is installed for you.
You have answered your own test prompt inside the real channel.
You have demonstrated to yourself that unpublished changes do not reach users.

If something goes wrong

Symptom	What to do
Publish fails or the button is unavailable	Most often a licensing issue — the trial licence cannot publish. Confirm your licence type.
The agent does not appear in Teams	The channel may not be connected, or the installation link has not been used. Also check that your Teams administrator permits custom agents.
Teams gives an old answer after publishing	Persistent channels keep the session. Type 'start over' in the chat, or wait for the refresh delay.
Organisation-wide publishing is blocked	This normally requires administrator approval in the Teams admin centre or Microsoft 365 admin centre.

Reflection

- What would you still need to do before letting a hundred real employees use this agent?

- Who should own this agent, and who should own each knowledge source?
- What would you measure in the first two weeks after launch?

Appendix A — Glossary

Terms you will meet on Day 1, in plain language.

Term	Meaning
Agent	An AI assistant that is given instructions, knowledge and tools, and decides for itself which steps to take to satisfy a request.
Agent flow	A deterministic, workflow-style automation an agent can call as a tool.
Channel	A place where users meet your agent — Teams, Microsoft 365 Copilot, a website, a phone line.
Citation	A reference under an answer showing which knowledge source was used.
Connector	A pre-built link to another system, used by tools and flows.
Copilot Credit	The unit of consumption billing across Copilot Studio, replacing 'messages' since September 2025.
Dataverse	The secure Microsoft data platform that stores your agent, its components and its transcripts.
DLP policy	A data loss prevention policy that controls which connectors may be used together in an environment.
Environment	A container in Power Platform holding data, apps, flows and agents.
Environment Maker	The standard Dataverse security role for someone who authors agents.
Generative answers	Answers composed by the model from your knowledge sources, rather than from a scripted topic.
Instructions	The text that defines an agent's identity, scope, tone, rules and fallback behaviour.
Knowledge source	Content the agent is allowed to read — files, websites, SharePoint, Dataverse, business systems, Microsoft IQ.
Managed environment	An environment with additional governance controls turned on by an administrator.
Microsoft IQ	The capability that gives an agent access to Microsoft 365 organisational data such as mail, calendar, files and Teams messages.
Orchestration	The runtime decision-making that chooses which knowledge, topic or tool answers a request.
Power Platform admin centre	Where administrators create environments, assign roles and apply policies.
Publish	The action that turns your edited version into the live version users see.
Skill	A reusable, self-contained set of instructions an agent can apply.
Solution	A package used to move components between environments.
Test canvas	The chat pane used while authoring. Called Preview in the new agent experience.
Tool	Something the agent can do — call an API, run a flow, use a custom prompt.
Topic	A designed conversation path with trigger phrases and nodes, used in the classic experience.

Appendix B — Day 1 Quick Reference

The one-page build checklist

1. Confirm you are in the right environment, and that it has Dataverse.
2. Confirm you hold the Environment Maker role.
3. Create the agent from a clear one-sentence description.
4. Rewrite the instructions using the six parts: Role, Scope, Sources, Tone and format, Fallback, Safety.
5. Add one authoritative knowledge source and wait for it to process.
6. Run the six-prompt test script and record the results.
7. Fix the weakest result by changing exactly one thing, then re-test.
8. Publish.
9. Connect the channel and set availability. Install for yourself first.
10. Confirm the answer in the real channel, not just in the test canvas.
11. Agree the owner, the refresh cycle for each source, and what you will measure in week one.

Six-part instruction template — copy this

Instruction template
Role: You are the _____________ assistant for _____________.
Scope: Answer only questions about _____________. Do not answer questions about _____________.
Sources: Use only _____________. Do not rely on general knowledge.
Tone and format: Be _____________. Answer in _____________ or fewer. Always _____________.
Fallback: If the answer is not available, say so clearly and direct the user to _____________.
Safety: Never _____________. Never _____________.

Six prompts to test every agent with

- A question you know the answer to.
- The same question in different words.
- A question that is in scope but not covered by your sources.
- A question that is clearly out of scope.
- An ambiguous question that should trigger a clarifying question.
- A nonsense or rude message.

Three things that break most first agents

Problem	Fix
Unclear instructions	Rewrite using the six-part template. Be positive and specific.
Too much, or stale, knowledge	One authoritative source, with an owner and a refresh cycle.
Forgetting to publish	Publish after every change and confirm in the channel.

Appendix C — Knowledge Check Answer Key

Module 1

Q	Answer	Why
1	B	A copilot assists a person who is driving; an agent is given a goal, decides its own steps, and can be started by an event.
2	C	The trial licence lets an individual create and test agents, but not publish them.
3	B	From 1 September 2025 the common currency changed from messages to Copilot Credits.
4	B	The new experience uses Build, Preview, Evaluate and Monitor. Topics, Knowledge, Actions and Settings belong to the classic experience.
5	C	There is no migration path in either direction; you choose the experience when you create the agent.

Module 2

Q	Answer	Why
1	B	An agent is stored in Dataverse, so the environment must have a Dataverse database.
2	C	Environment Maker is the standard maker role, and is also required for co-authors.
3	B	Analytics Viewer grants access to a specific agent's Analytics page without edit rights.
4	B	In the default environment every tenant user automatically holds Environment Maker and it cannot be removed.
5	B	Changes reach users only after a publish; persistent channels may also need the session to reset.

Module 3

Q	Answer	Why
1	B	The Build tab consolidates instructions, knowledge, tools, skills and model.
2	C	The fallback tells the agent what to do when it cannot find an answer.
3	B	No citation usually means the answer was not grounded — check the source is attached and finished processing.
4	B	SharePoint knowledge respects each user's own permissions, so answers legitimately differ.
5	B	Instructions first, one knowledge source, test, then extend.

Appendix D — Notes and Next Steps

Use these pages for your own notes, questions for the trainer, and ideas for agents in your own team.

Three agents I could build for my own team

#	Agent purpose (one sentence)	Who uses it	First knowledge source
1	______________________________	_______________	_______________
2	______________________________	_______________	_______________
3	______________________________	_______________	_______________

Coming up on Day 2

- Topics, trigger phrases and conversation design.
- Variables, entities and slot filling.
- Actions and tools — connecting your agent to real systems.
- Advanced conversation patterns and escalation to a human.
- Analytics, evaluation and continuous improvement.